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Data Visualization

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detail

ÿ seaborn

- Data visualization
- Creation of charts

ÿ matplotlib

- Chart creation method
- Utilizing seaborn

ÿ Visualization Practice

- Data preparation
- Chart creation practice

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seaborn

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Visualization of data (1)

ÿ Purpose and Advantages of Data Visualization

- Advantage of enabling the discovery of inherent patterns while conveying a large amount of information

ÿ Packages providing visualization tools for Python: seaborn, matplotlib

- seaborn: Based on matplotlib, with added color themes and chart features

  high-level interface

  - Use via import: import seaborn as sns

- matplotlib: The most basic package for data visualization

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Visualization of data (2)

ÿ Verify functionality using datasets provided by the seaborn package

- Dataset name: Called via sns.get_dataset_names()
  - 'anagrams', ... , 'iris', 'mpg', 'penguins', 'planets', 'tips', 'titanic', etc.

ÿ Explanation of visualization forms using seaborn through example graphs

- Scatterplot, regression line, lineplot, bar graph

  Raff (barplot, countplot), boxplot, violinplot,

  Histograms (histplots), heatmaps, etc.

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Creating a chart: dataset

- Import seaborn package
  - import seaborn as sns
    - Usually, the namespace is set to 'sns' for use.
- Datasets provided by seaborn
  - Provided by the seaborn package

    Check dataset:

    call get_dataset_names()

```
In [1]: import seaborn as sns
        sns.get_dataset_names()
```

```
Out[1]: ['anagrams',
         'anscombe',
         'attention',
         'brain_networks',
         'car_crashes',
         'diamonds',
         'dots',
         'exercise',
         'flights',
         'fmri',
         'gammas',
         'geyser',
         'iris',
         'mpg',
         'penguins',
         'planets',
         'taxis',
         'tips',
         'titanic']
```

- "tips" dataset regarding tips paid at restaurants, amounts, customer information, etc.
  - Reads data by calling load_dataset() and returns it in the form of a DataFrame.

```
In [2]: import seaborn as sns
        import pandas as pd
        df = sns.load_dataset("tips")
        df.head()
```

Out[2]:

	total_bill	tip	sex	smoker	day	time	
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

- Data attributes

total_bill	tip	sex	smoker	day	time	size
전체 금액	팁 금액	성별	흡연여부	방문요일	방문시간	인원 수

  - Check data information

```
In [3]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   total_bill  244 non-null    float64
 1   tip         244 non-null    float64
 2   sex         244 non-null    category
 3   smoker      244 non-null    category
 4   day         244 non-null    category
 5   time        244 non-null    category
 6   size        244 non-null    int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
```

  - Check the distribution of numeric data values using describe()

```
In [4]: df.describe()
```

Out[4]:

	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

- etc

```
In [5]: df['sex'].unique()
Out[5]: ['Female', 'Male']
        Categories (2, object): ['Female', 'Male']
In [6]: df['sex'].value_counts()
Out[6]: Male      157
        Female     87
        Name: sex, dtype: int64
```

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Creating a plot: scatterplot

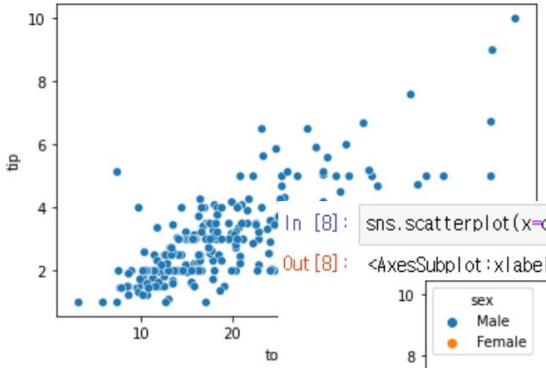
- Useful for checking the correlation between x-axis and y-axis variable values

sns.*scatterplot*(x, y)

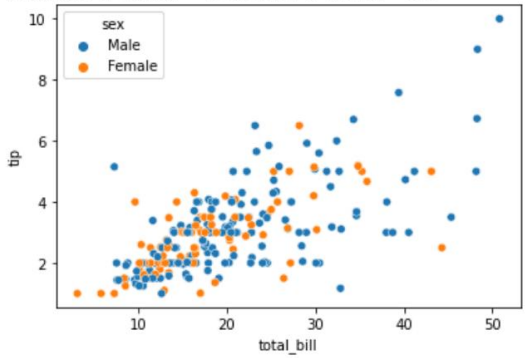
x	x축 변수
y	y 축 변수
hue	그룹핑할 변수(색상)
size	그룹핑할 변수(점의 크기)
style	그룹핑할 변수(점의 모양)
data	데이터프레임 또는 다차원 배열

- example

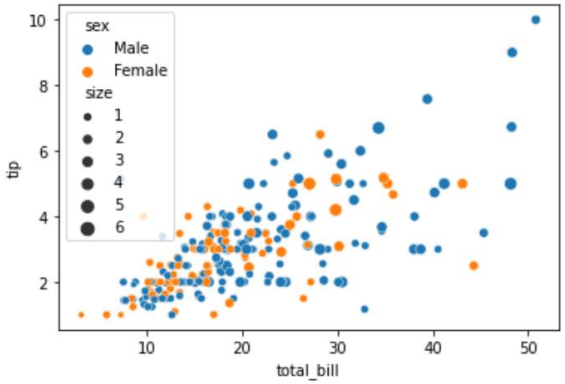
```
In [7]: sns.scatterplot(x=df['total_bill'], y=df['tip'])
Out[7]: <AxesSubplot:xlabel='total_bill', ylabel='tip'>
```




```
In [8]: sns.scatterplot(x=df['total_bill'], y=df['tip'], hue=df['sex'])
Out[8]: <AxesSubplot:xlabel='total_bill', ylabel='tip'>
```

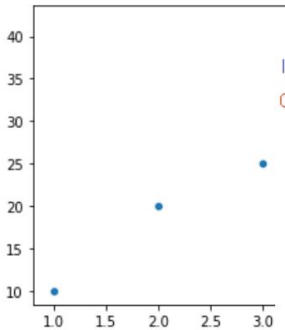



```
In [9]: sns.scatterplot(x=df['total_bill'], y=df['tip'], hue=df['sex'],
        size=df['size'] )
Out[9]: <AxesSubplot:xlabel='total_bill', ylabel='tip'>
```

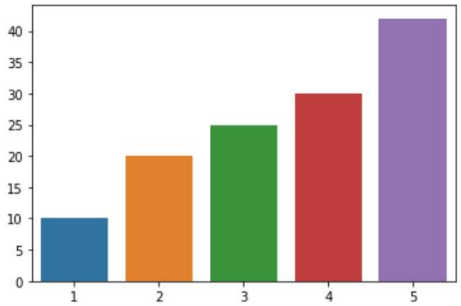


- Visualize with scatter plots and bar graphs when there is little data to display

```
In [10]: x= [1, 2, 3, 4, 5]
         y= [10, 20, 25, 30, 42]
         sns.scatterplot(x=x, y=y)
Out[10]: <AxesSubplot:>
```



```
In [11]: sns.barplot(x=x, y=y)
Out[11]: <AxesSubplot:>
```



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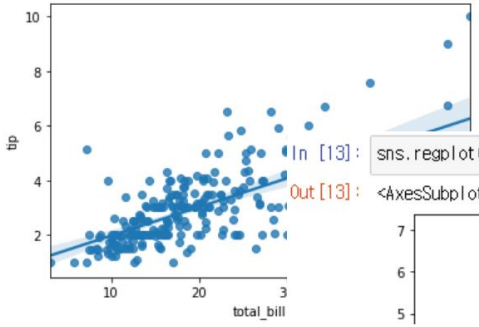
Drawing charts: regression line, line graph

- A graph displaying a regression line

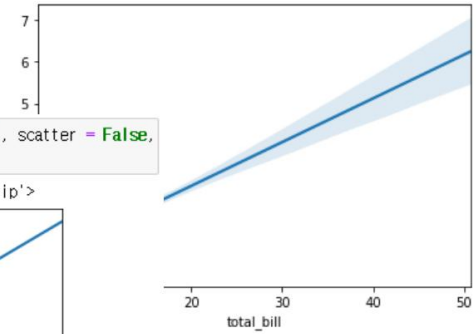
sns.*regplot*(x, y)	
x	x축 변수
y	y축 변수
scatter	산점도(scatterplot)를 함께 표시할지 여부
ci	신뢰구간(기본: 95%)
data	데이터프레임 또는 다차원 배열

- example

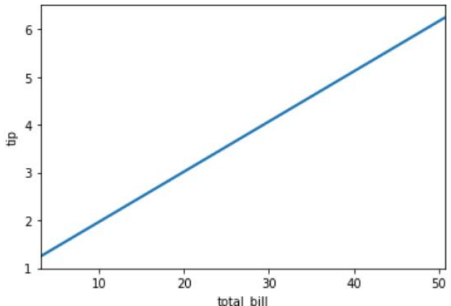
```
In [12]: sns.regplot(x=df['total_bill'], y=df['tip'])
Out[12]: <AxesSubplot:xlabel='total_bill', ylabel='tip'>
```




```
In [13]: sns.regplot(x=df['total_bill'], y=df['tip'], scatter = False)
Out[13]: <AxesSubplot:xlabel='total_bill', ylabel='tip'>
```




```
In [14]: sns.regplot(x=df['total_bill'], y=df['tip'], scatter = False,
                     ci = None)
Out[14]: <AxesSubplot:xlabel='total_bill', ylabel='tip'>
```

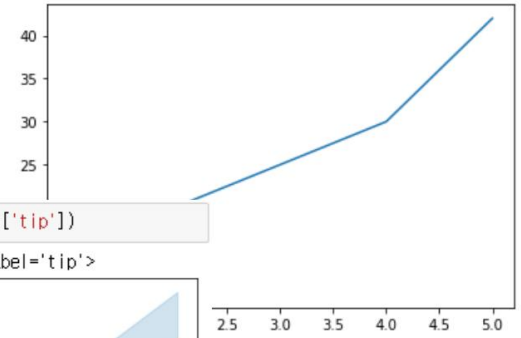


- Use seaborn's lineplot for the line graph

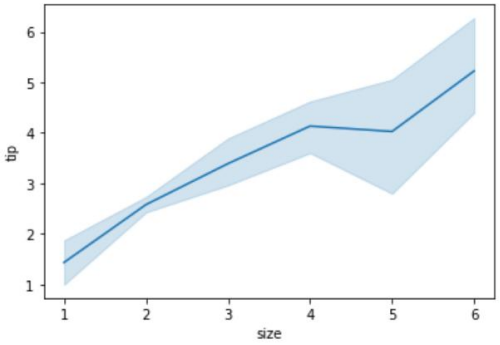
sns.*lineplot*(x, y)	
x	x축 변수
y	y축 변수
hue	그룹핑할 변수(색상)
size	그룹핑할 변수(점의 크기)
style	그룹핑할 변수(점의 모양)
data	데이터프레임 또는 다차원 배열

- example

```
In [15]: x= [1, 2, 3, 4, 5]
         y= [10, 20, 25, 30, 42]
         sns.lineplot(x=x, y=y)
Out[15]: <AxesSubplot:>
```



```
In [16]: sns.lineplot(x=df['size'], y=df['tip'])
Out[16]: <AxesSubplot:xlabel='size', ylabel='tip'>
```



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Creating a chart: line graph

- Change in confidence interval

  – Background in lineplot: Confidence interval of the value

  - Default: Based on a 95% confidence interval -

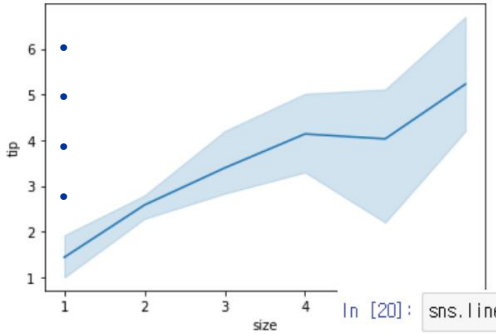
  Change confidence interval: Change CI value

```
In [18]: print("min:{}, max:{}".format(min(df[df['size'] == 5]['tip']),
         max(df[df['size'] == 5]['tip'])))

min:2.0, max:5.14
```

  - When size is 5, tip minimum and maximum values: 2.0, 5.14

```
In [19]: sns.lineplot(x=df['size'], y=df['tip'], ci=100)
Out[19]: <AxesSubplot:xlabel='size', ylabel='tip'>
```



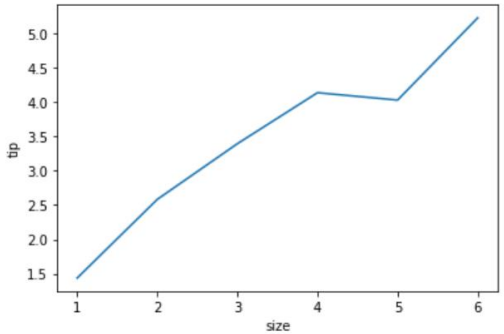
All data

mark:

ci = 100

setting

```
In [20]: sns.lineplot(x=df['size'], y=df['tip'], ci=None)
Out[20]: <AxesSubplot:xlabel='size', ylabel='tip'>
```

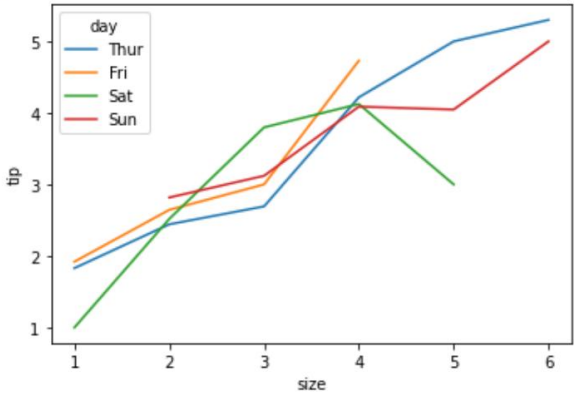


No confidence interval displayed:

Set ci = None

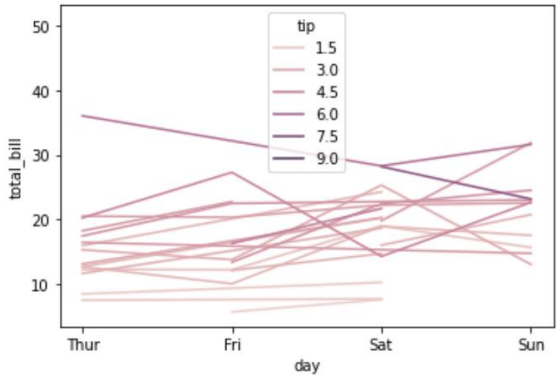
- hue option, display different colors

```
In [21]: sns.lineplot(x=df['size'], y=df['tip'], hue=df['day'], ci=None)
Out[21]: <AxesSubplot:xlabel='size', ylabel='tip'>
```



– The hue property specifies categorical variables – Specifying numerical variables: graphs that are difficult to interpret

```
In [22]: sns.lineplot(x=df['day'], y=df['total_bill'], hue=df['tip'], ci=None)
Out[22]: <AxesSubplot:xlabel='day', ylabel='total_bill'>
```



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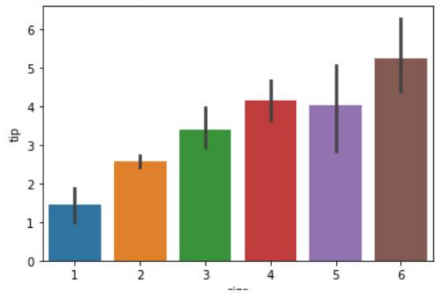
Creating charts: Bar graphs (barplot, countplot)

- Using seaborn's barplot

sns.*barplot*(x, y)	
x	x축 변수
y	y축 변수
hue	그룹핑할 변수(색상)
orient	막대의 방향(기본: 'vertical', 수평: 'horizontal')
ci	신뢰구간(기본: 95%)
data	데이터프레임 또는 다차원 배열

- example

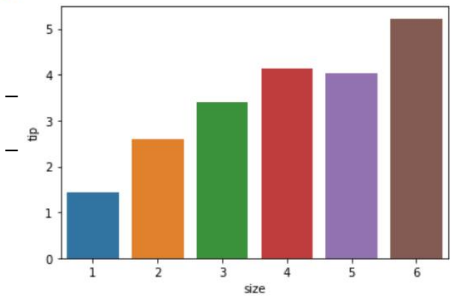
```
In [23]: sns.barplot(x=df['size'], y=df['tip'])
Out[23]: <AxesSubplot:xlabel='size', ylabel='tip'>
```



Bars of a bar graph

Centerline: Confidence Interval

```
In [24]: sns.barplot(x=df['size'], y=df['tip'], ci=None)
Out[24]: <AxesSubplot:xlabel='size', ylabel='tip'>
```

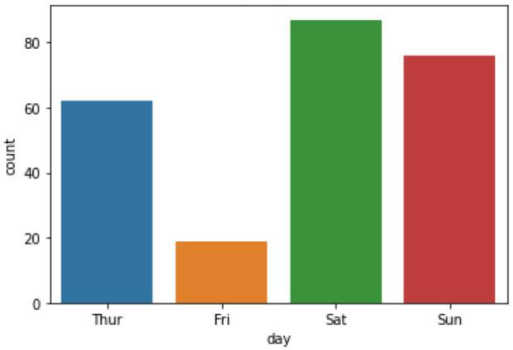


without a centerline

Bar graph: ci=None

- countplot() similar to barplot()

```
In [25]: sns.countplot(x=df['day'])
Out[25]: <AxesSubplot:xlabel='day', ylabel='count'>
```



The y-axis value is the x-axis

Number of variables

(count) display

(to set on the x-axis

(Specify only variables)

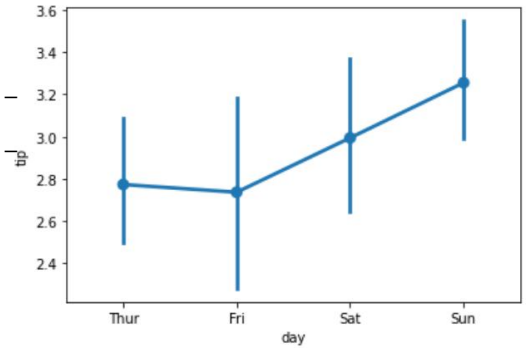
– countplot() is the result of value_counts() of the dataframe

Visualize

```
In [26]: df['day'].value_counts()
Out[26]: Sat     87
         Sun     76
         Thur    62
         Fri     19
         Name: day, dtype: int64
```

– pointplot

```
In [27]: sns.pointplot(x=df['day'], y=df['tip'])
Out[27]: <AxesSubplot:xlabel='day', ylabel='tip'>
```



bars in a bar graph

Exclusion, average of the data

Connect the values with a line

Display the confidence interval together

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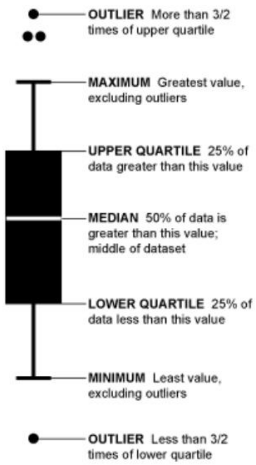
Diagram Construction: Box Plot, Violin Plot

- Boxplot indicator lines



– Minimum value: The position 1.5 IQR below the first quartile

– 1st Quartile (Q1): 25th percentile position

– 2nd Quartile (Q2): 50th percentile position, median – 3rd Quartile

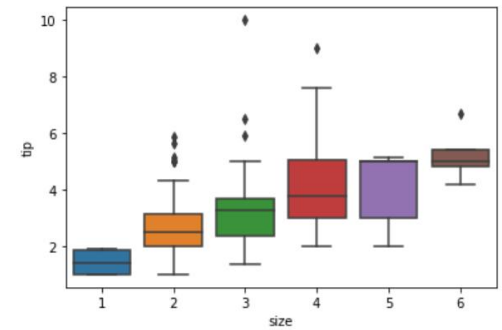
(Q3): 75th percentile position – Maximum: 1.5 IQR added to the 3rd quartile

- Using seaborn's boxplot

sns.*boxplot*(x, y)	
x	x축 변수
y	y축 변수
hue	그룹핑할 변수(색상)
orient	막대의 방향(기본: 'vertical', 수평: 'horizontal')
data	데이터프레임 또는 다차원 배열

- Boxplot examples

```
In [28]: sns.boxplot(x=df['size'], y=df['tip'])
Out[28]: <AxesSubplot:xlabel='size', ylabel='tip'>
```

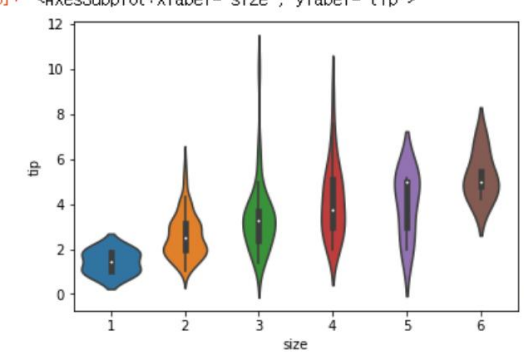


- Using seaborn's violinplot

sns.*violinplot*(x, y)	
x	x축 변수
y	y축 변수
hue	그룹핑할 변수(색상)
orient	막대의 방향(기본: 'vertical', 수평: 'horizontal')
data	데이터프레임 또는 다차원 배열

- example

```
In [29]: sns.violinplot(x=df['size'], y=df['tip'])
Out[29]: <AxesSubplot:xlabel='size', ylabel='tip'>
```



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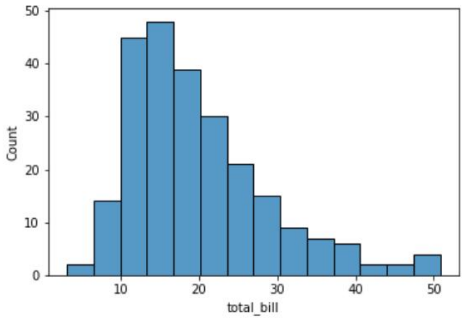
Plot Creation: Histogram

- Using seaborn's histplot

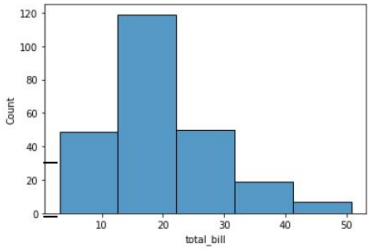
sns.*histplot*(data)	
data	데이터프레임 또는 다차원 배열
hue	그룹핑할 변수(색상)
bins	막대의 개수
kde	막대를 부드럽게 연결한 선을 표시할지 여부(kernel density estimation, 커널 밀도 추정)

- example

```
In [30]: sns.histplot(data=df['total_bill'])
Out[30]: <AxesSubplot:xlabel='total_bill', ylabel='Count'>
```



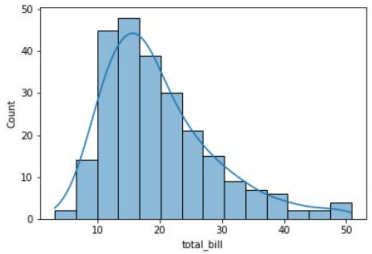

```
In [31]: sns.histplot(data=df['total_bill'], bins = 5)
Out[31]: <AxesSubplot:xlabel='total_bill', ylabel='Count'>
```



Set the number of bars:

bins attribute

```
In [32]: sns.histplot(data=df['total_bill'], kde =True)
Out[32]: <AxesSubplot:xlabel='total_bill', ylabel='Count'>
```



Data density

(Kernel Density

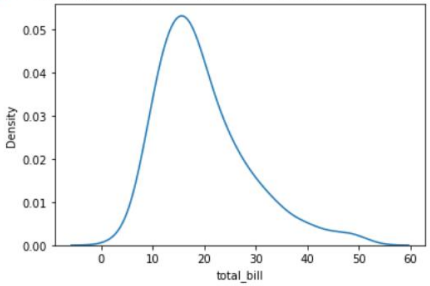
Estimation) Display:

kde attributes

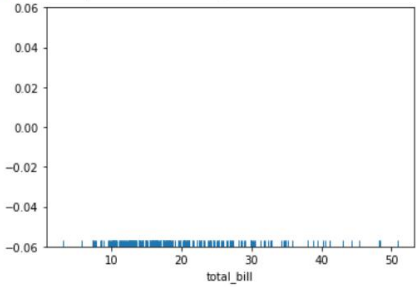
Displayed as a curve

- kdeplot and rugplot

```
In [33]: sns.kdeplot(data=df['total_bill'])
Out[33]: <AxesSubplot:xlabel='total_bill', ylabel='Density'>
```




```
In [34]: sns.rugplot(data=df['total_bill'])
Out[34]: <AxesSubplot:xlabel='total_bill'>
```



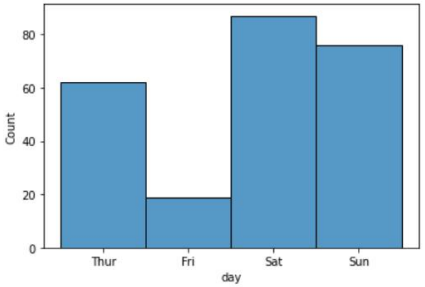
- Histograms check the distribution of continuous variables

  graph used when doing

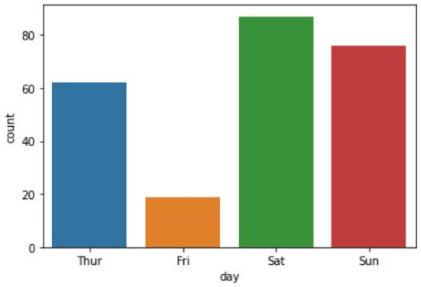
  – When using categorical data,

  Similar to a countplot

```
In [35]: sns.histplot(data=df['day'])
Out[35]: <AxesSubplot:xlabel='day', ylabel='Count'>
```




```
In [36]: sns.countplot(x=df['day'])
Out[36]: <AxesSubplot:xlabel='day', ylabel='count'>
```



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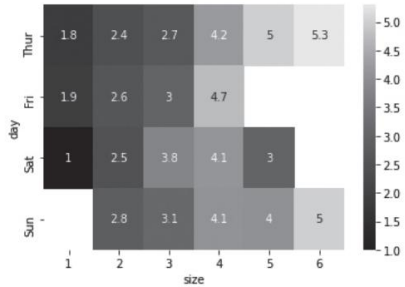
Creating a chart: heatmap

- Using seaborn's heatmap()

sns.*heatmap*(data)	
data	2D 데이터셋(rectangular dataset)
annot	값 표시 여부

- Heatmap displaying the average tip payment amount by size and day



– Mean, Max, Sum, Min, Count, etc.

- Generating data containing the data needed for the heatmap: Created by pivoting the dataframe

```
In [37]: pivot_df = df.pivot_table("tip", "day", "size")
         pivot_df
```

Out [37]:

size / day	1	2	3	4	5	6
Thur	1.83	2.442500	2.692500	4.218000	5.000000	5.3
Fri	1.92	2.644375	3.000000	4.730000	NaN	NaN
Sat	1.00	2.517547	3.797778	4.123846	3.000000	NaN
Sun	NaN	2.816923	3.120667	4.087778	4.046667	5.0

- Changes to pivot table value operations

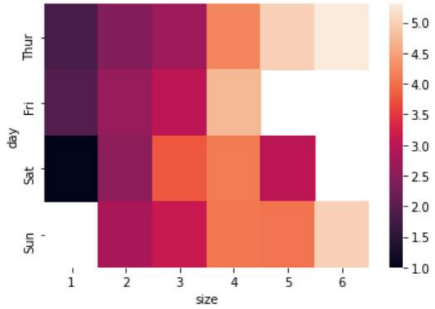
```
In [38]: df.pivot_table("tip", "day", "size", aggfunc='count')
```

Out [38]:

size / day	1	2	3	4	5	6
Thur	1	48	4	5	1	3
Fri	1	16	1	1	0	0
Sat	2	53	18	13	1	0
Sun	0	39	15	18	3	1

Change by specifying the aggfunc property (specify the pivot table value as the co

```
In [39]: sns.heatmap(pivot_df)
Out [39]: <AxesSubplot:xlabel='size', ylabel='day'>
```



Set missing values to 0

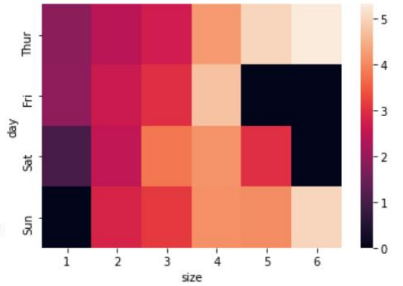
conversion

```
In [40]: pivot_df.fillna(0, inplace=True)
         pivot_df
```

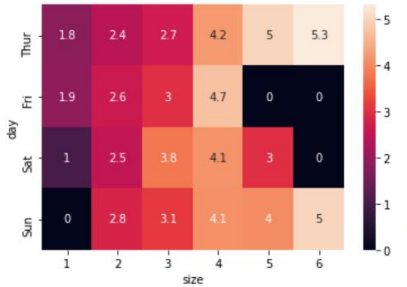
Out [40]:

size / day	1	2	3			
Thur	1.83	2.442500	2.692500	4.218000	5.000000	5.3
Fri	1.92	2.644375	3.000000	4.730000	0.000000	0.0
Sat	1.00	2.517547	3.797778	4.123846	3.000000	0.0
Sun	0.00	2.816923	3.120667	4.087778	4.046667	5.0

```
In [41]: sns.heatmap(pivot_df)
Out [41]: <AxesSubplot:xlabel='size', ylabel='day'>
```




```
In [42]: sns.heatmap(pivot_df, annot=True)
Out [42]: <AxesSubplot:xlabel='size', ylabel='day'>
```



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matplotlib

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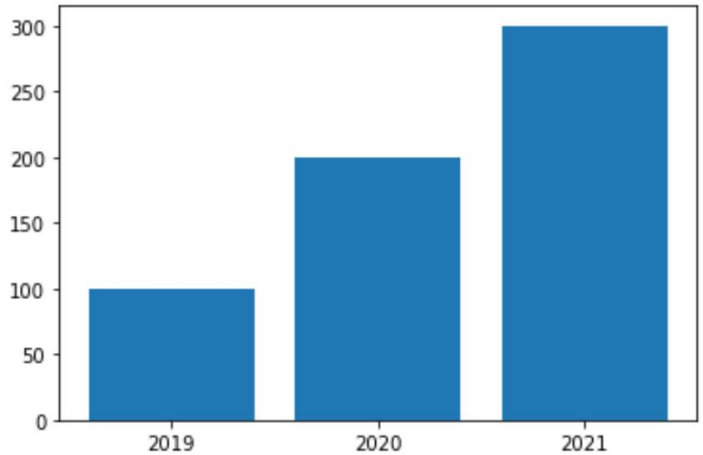
Chart creation method (1)

• Configuration Examples

```
In [43]: import matplotlib.pyplot as plt
         x = ['2019', '2020', '2021']
         y = [100, 200, 300]
         plt.bar(x,y)
         plt.show()
```



– Seaborn displays a bar graph with a single line of code

– matplotlib generates bar graphs, and

Execute the command to show

• plt.bar() function parameters

width	막대 너비
align	x축 시작 위치(기본값 = center)
color	막대 색상
edgecolor	선 색상
linewidth	선 너비

• example

```
In [44]: x = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]
         y = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120]
         plt.bar(x, y, width=1, align='edge', color="red", edgecolor="blue",
                 linewidth=2)
         plt.show()
```

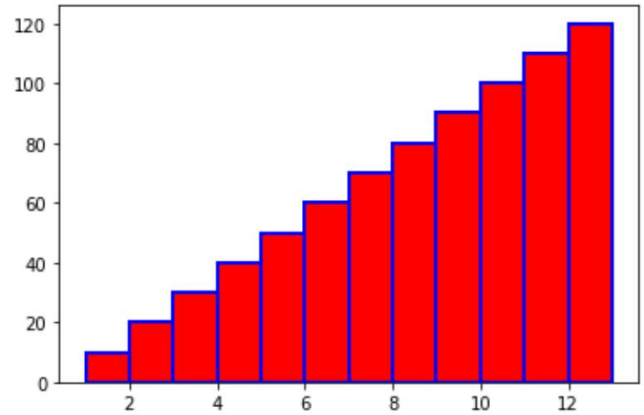


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Chart creation method (2)

ÿ Figure and plot

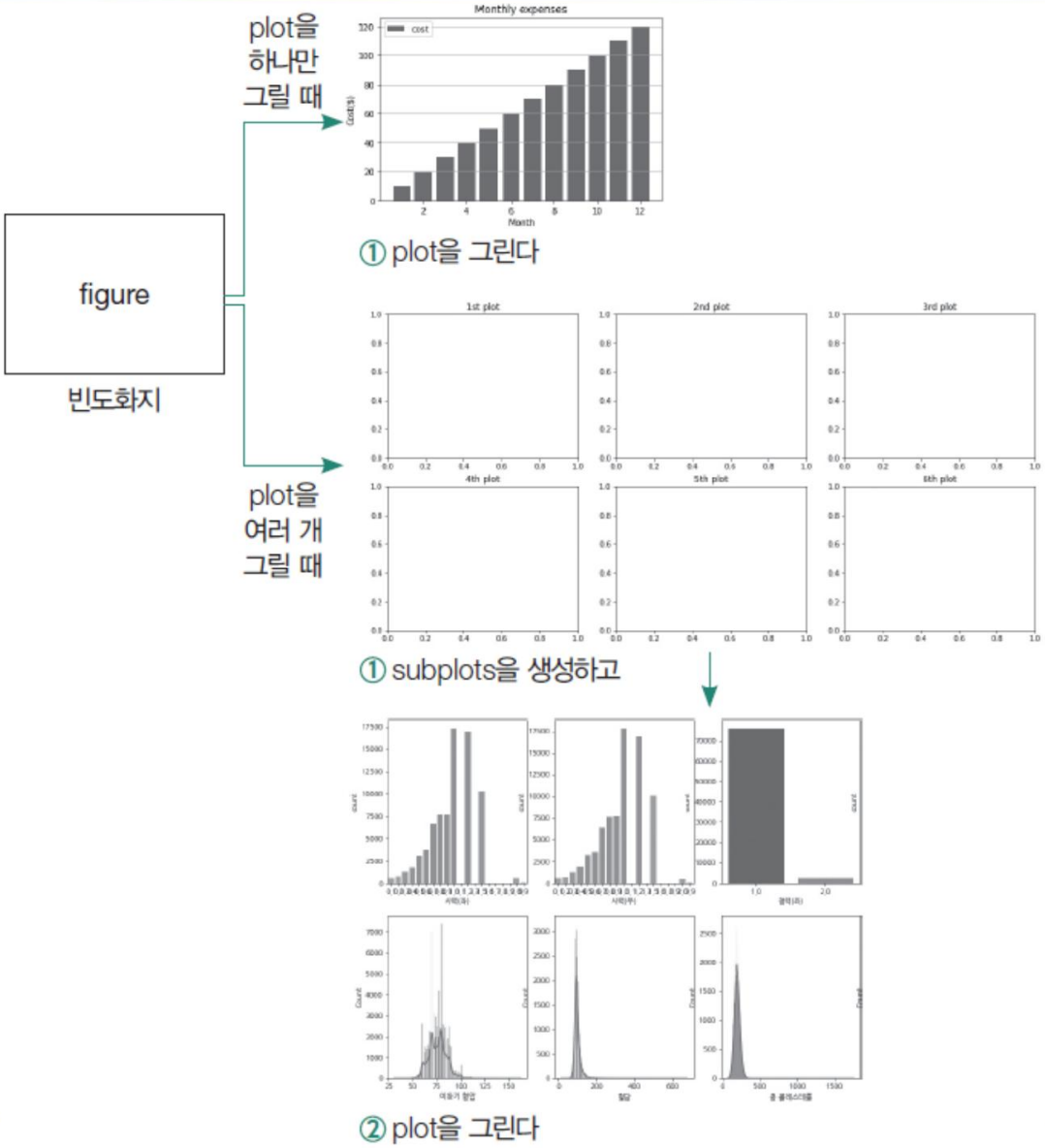

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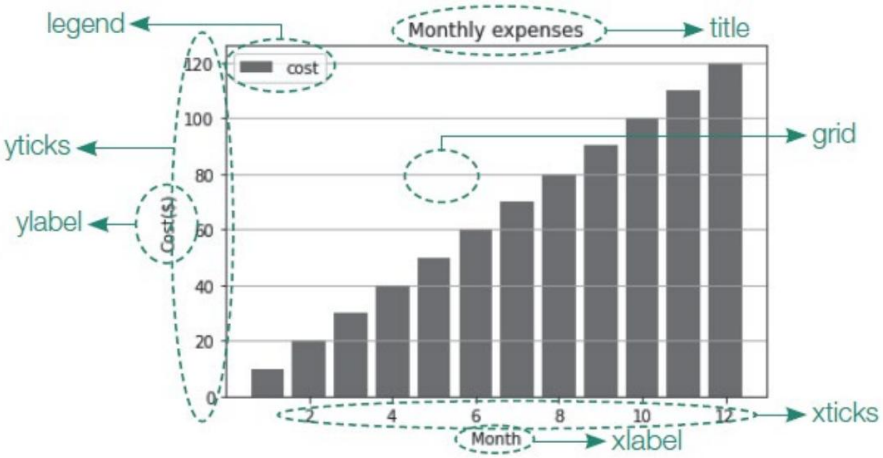
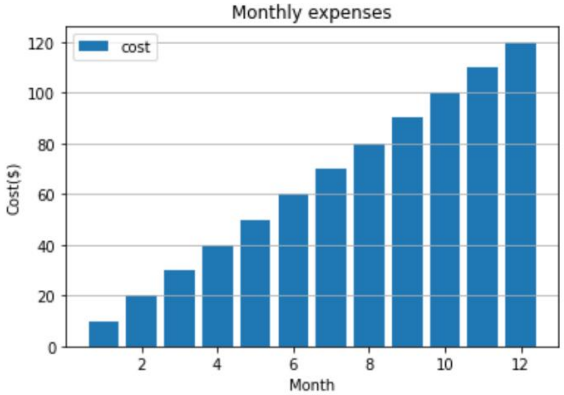
Chart creation method (3)

ÿ Set plots on figure



ÿ Case

In [45]:
```
x = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]
y = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120]
plt.bar(x, y)
# 추가 ----------------------
plt.title('Monthly expenses') #제목
plt.xlabel('Month') #x축 이름
plt.ylabel('Cost($)') #y축 이름
plt.legend(['cost']) #범례
plt.grid(True, axis='y') #눈금
#----------------------------
plt.show()
```



In [46]:
```
x = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]
y = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120]
# 추가
fig = plt.figure(figsize=(10,5)) # 가로, 세로 크기.
plt.bar(x, y)
plt.title('Monthly expenses')
plt.xlabel('Month')
plt.ylabel('Cost($)')
plt.grid(True, axis='y')
plt.show()
```

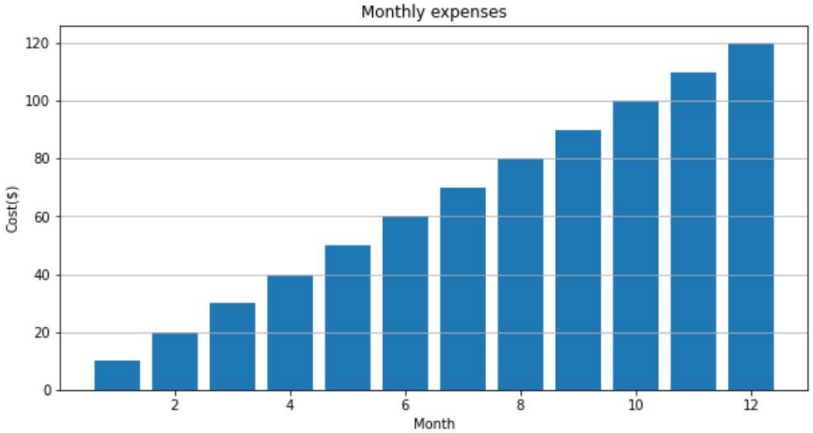


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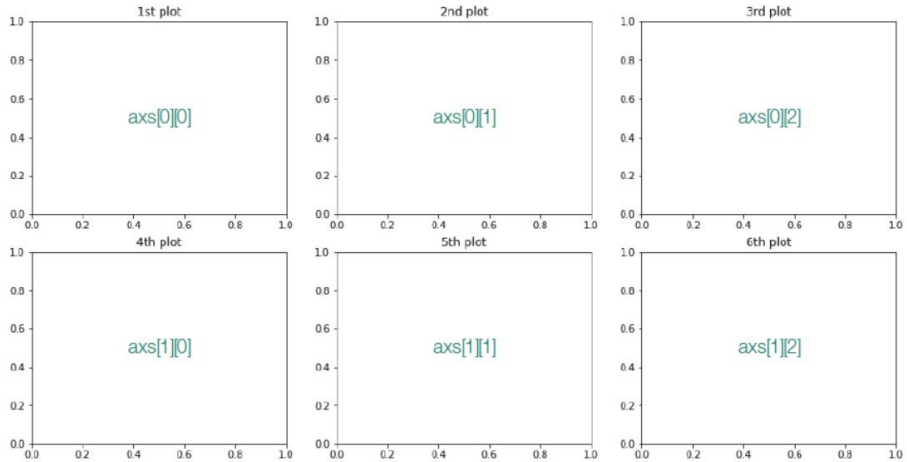
Chart creation method (4)

ÿ Divide a figure into multiple sections: Use subplots

- fig, axs = plt.subplots(nrows=2, ncols=3)



- fig, ((ax1,ax2,ax3), (ax4,ax5,ax6)) = plt.subplots(nrows=2, ncols=3)

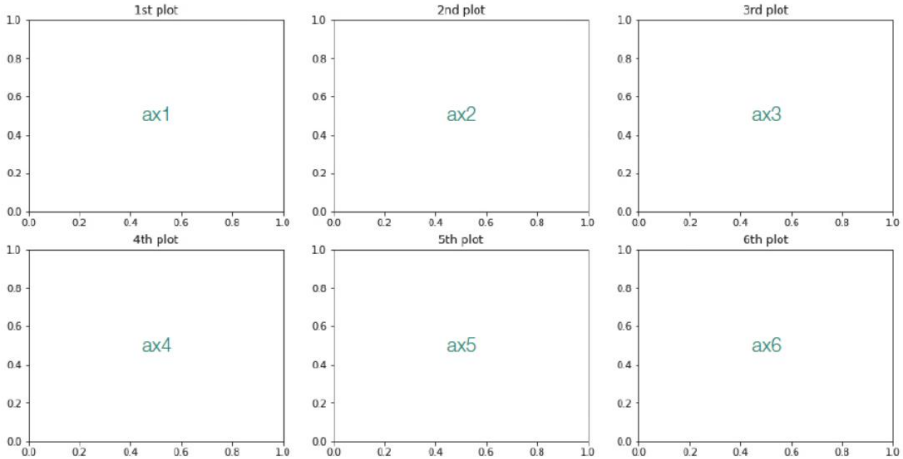


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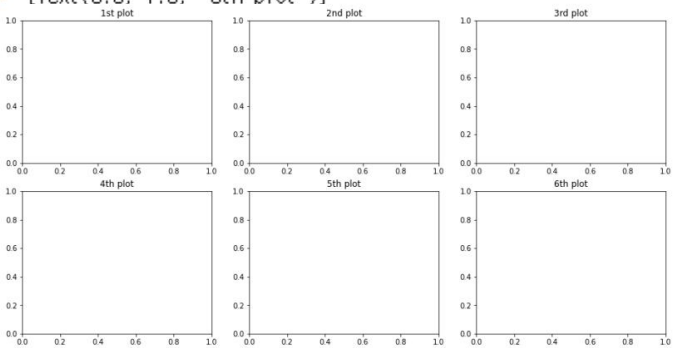
Chart creation method (5)

ÿ Use cases for subplots

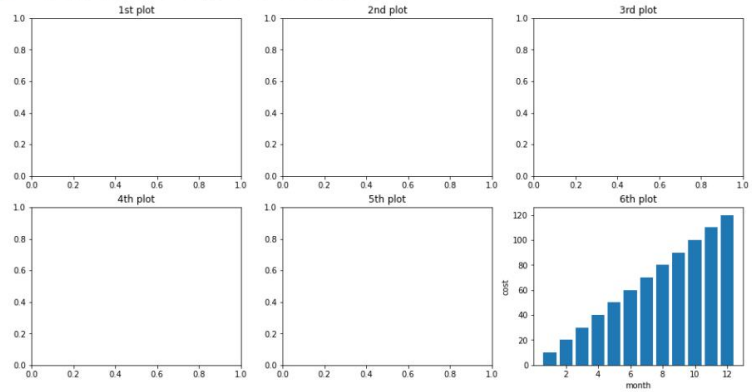
In [47]:
```
fig, ((ax1,ax2,ax3), (ax4,ax5,ax6)) = plt.subplots(nrows=2,
ncols=3) #여섯 개의 도화지 생성
fig.set_size_inches(16,8) #사이즈 설정
#각각의 도화지 이름 설정
ax1.set(title="1st plot")
ax2.set(title="2nd plot")
ax3.set(title="3rd plot")
ax4.set(title="4th plot")
ax5.set(title="5th plot")
ax6.set(title="6th plot")
```

Out[47]: [Text(0.5, 1.0, '6th plot')]



In [48]:
```
fig, ((ax1,ax2,ax3), (ax4,ax5,ax6)) = plt.subplots(nrows=2, ncols=3)
fig.set_size_inches(16,8)
ax1.set(title="1st plot")
ax2.set(title="2nd plot")
ax3.set(title="3rd plot")
ax4.set(title="4th plot")
ax5.set(title="5th plot")
ax6.set(xlabel='month', ylabel='cost', title="6th plot")
ax6.bar(x,y)
```

Out[48]: <BarContainer object of 12 artists>



ÿ Minor problems

- Issue with broken Korean characters: Set to a font that supports Korean

  (Example) import matplotlib.pyplot as plt plt.rcParams["font.family"] = 'NanumGothic'

- Issue where the minus sign is not displayed: Change the rcParams setting

  (Example) import matplotlib as mpl

  import matplotlib.pyplot as plt

  mpl.rcParams['axes.unicode_minus'] = False • Display values on

the axis in an italicized manner: (e.g.) plt.xticks(rotation = 45)

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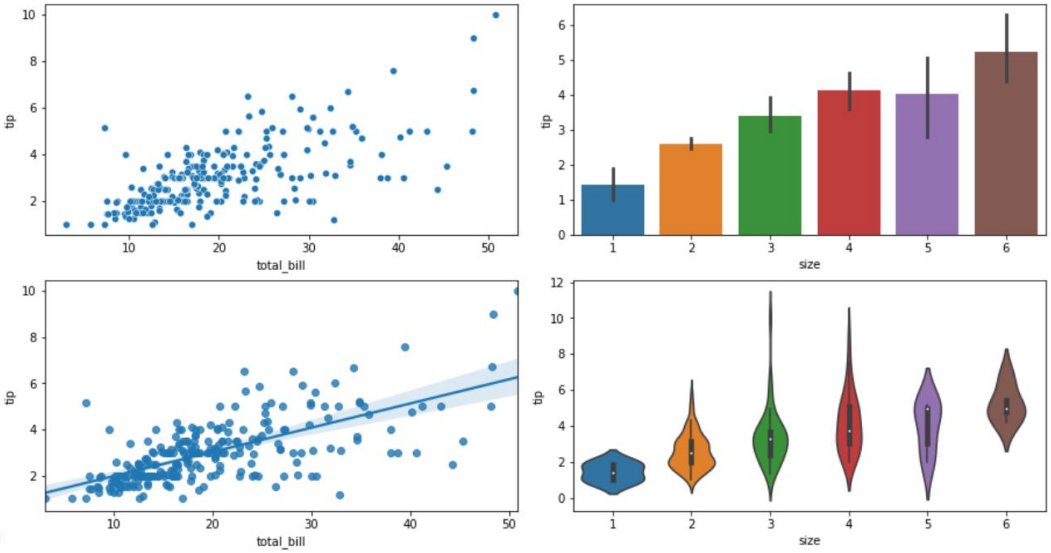
Using seaborn (1)

ÿ Use case of seaborn and matplotlib in parallel (1)

In [49]:
```
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt

df = sns.load_dataset("tips")

fig, ax = plt.subplots(nrows=2, ncols=2)

fig.set_size_inches(16,8)

sns.scatterplot(x=df["total_bill"], y=df["tip"], ax=ax[0][0])
sns.barplot(x=df['size'], y=df['tip'], ax=ax[0][1])
sns.regplot(x=df["total_bill"], y=df["tip"], ax=ax[1][0])
sns.violinplot(x=df['size'], y=df['tip'], ax=ax[1][1])
```

Out [49]:
```
<AxesSubplot:xlabel='size', ylabel='tip'>
```



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Using seaborn (2)

ÿ Concurrent use of aborn and matplotlib (2)

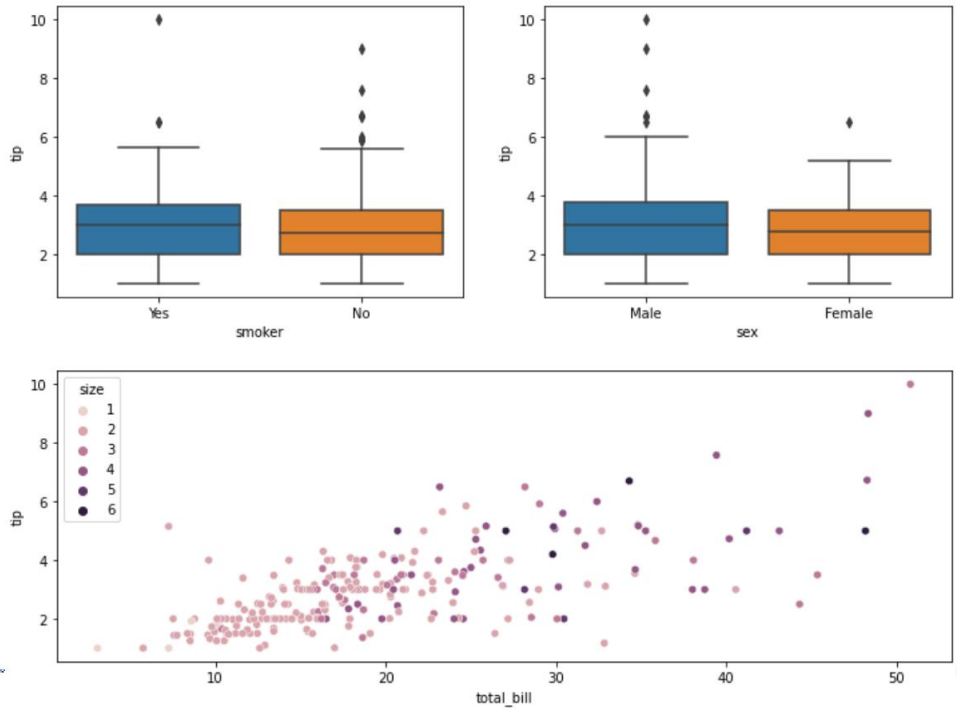
In [50]:
```
fig, (ax1, ax2) = plt.subplots(nrows=1, ncols=2)
fig.set_size_inches(12, 4)

sns.boxplot(x=df["smoker"], y=df["tip"], ax=ax1)
sns.boxplot(x=df["sex"], y=df["tip"], ax=ax2)

fig, ax3 = plt.subplots(nrows=1, ncols=1)
fig.set_size_inches(12, 4)

sns.scatterplot(x=df["total_bill"], y=df["tip"], hue=df["size"], ax=ax3)
```

Out [50]:
```
<AxesSubplot:xlabel='total_bill', ylabel='tip'>
```



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Visualization practice

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Data preparation (1)

ÿ National Health Insurance Service health checkup information data (data.go.kr)

- Download National Health Insurance Service health checkup data
  - National Health Insurance Service_HealthCheckupInformation_20201231_5.csv
- Read as a DataFrame. • Set

encoding (euc-kr) for Korean processing. • National

Health Insurance Service_HealthCheckupInformation_20201231_5.csv

In [51]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
plt.rcParams["font.family"] = 'NanumGothic'
df = pd.read_csv("국민건강보험공단_건강검진정보_20201231_5.csv", encoding = 'euckr')
df.head()
```

Out [51]:

	기준년도	시도코드	성별코드	연령대코드(5세단위)	신장(5Cm단위)	체중(5Kg단위)	허리둘레	시력(좌)	시력(우)	청력(좌)	...	혈청크레아티닌	(혈청지오티)AST	(혈청지오티)ALT	감마지티피	흡연상태	음주여부	구강검진수검여부	치아우식증유무	치석	데이터 공개일자
0	2020	11	1.0	10.0	175.0	65.0	92.0	1.0	1.0	1.0	...	0.8	20.0	22.0	25.0	1.0	1.0	0	NaN	NaN	20210929
1	2020	11	2.0	15.0	150.0	60.0	85.0	0.5	1.0	1.0	...	0.6	35.0	22.0	16.0	1.0	NaN	0	NaN	NaN	20210929
2	2020	41	2.0	15.0	155.0	60.0	80.0	0.4	0.5	1.0	...	0.7	29.0	22.0	23.0	1.0	NaN	0	NaN	NaN	20210929
3	2020	41	1.0	9.0	165.0	60.0	83.0	1.0	0.7	1.0	...	0.9	22.0	22.0	17.0	2.0	NaN	1	0.0	0.0	20210929
4	2020	31	1.0	9.0	160.0	55.0	78.0	1.2	1.2	1.0	...	1.0	27.0	22.0	20.0	2.0	1.0	0	NaN	NaN	20210929

5 rows × 30 columns

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Data preparation (2)

ÿ Required preprocessing

- Delete columns unnecessary for data analysis (drop) •

Rename columns with long names •

Remove rows containing missing values (dropna), etc.

```
In [56]: df.drop(['기준년도', '치아우식증유무', '데이터 공개일자'], inplace=True, axis = 1)
         df.rename(columns = {'연령대 코드(5세단위)':'연령코드', '신장(5Cm단위)':'신장',
                              '체중(5Kg 단위)': '체중', '식전혈당(공복혈당)': '혈당' }, inplace = True)
         df = df.dropna()
```

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Chart creation practice (1)

ÿ Construct the data from 25 columns sequentially into 5x5 subplots

Visualization on the intact figure

- Histogram (histplot) and countplot
- Column attribute values are less than 30 types

  In this case, a countplot; otherwise, a histogram

```
fig, axs = plt.subplots(5, 5)
fig.set_size_inches(20,24)
for i in range(5):
    for j in range(5):
        attr = i * 5 + j + 1
        if df[df.columns[attr]].nunique() < 30:
            sns.countplot( x=df.columns[attr], data=df,ax=axs[i][j])
        else:
            sns.histplot( x=df.columns[attr], data=df, kde=True, ax=axs[i][j])
```

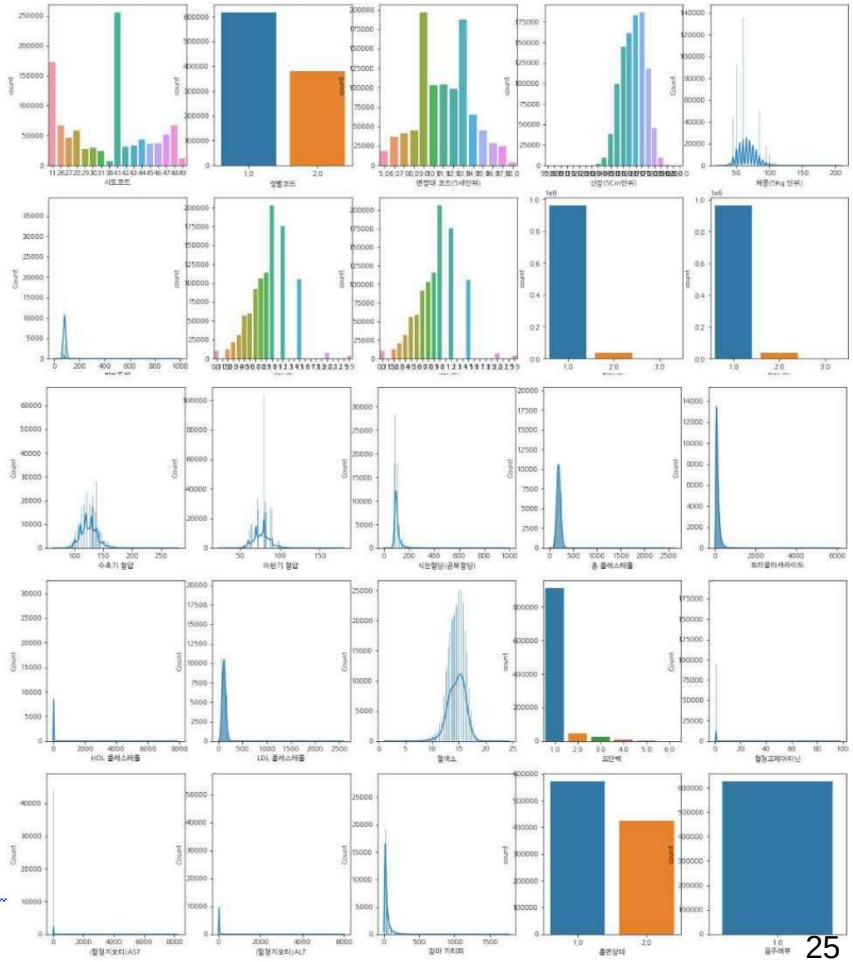

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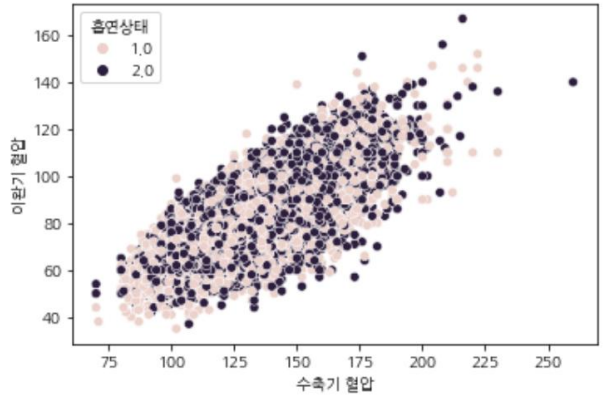
Chart creation practice (2)

ÿ Visual analysis and verification of various attribute values, such as trends, distributions, differences, and correlations between attributes

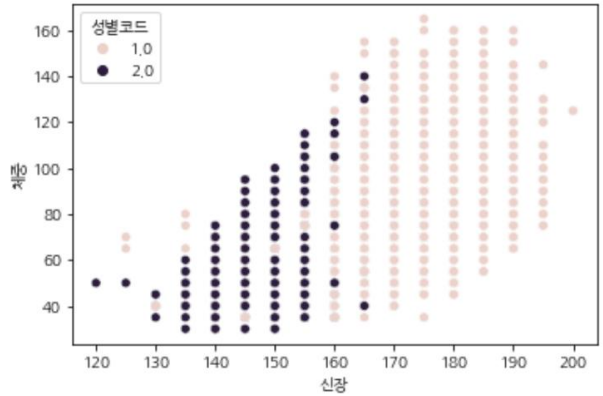
In [58]:
```
sns.scatterplot(x=df['수축기 혈압'], y=df['이완기 혈압'], hue=df['흡연상태'])
```

Out [58]:
```
<AxesSubplot:xlabel='수축기 혈압', ylabel='이완기 혈압'>
```



In [59]:
```
sns.scatterplot(x=df['신장'], y=df['체중'], hue=df['성별코드'])
```

Out [59]:
```
<AxesSubplot:xlabel='신장', ylabel='체중'>
```



In [60]:
```
sns.scatterplot(x=df['혈당'], y=df['총 콜레스테롤'], hue=df['성별코드'])
```

Out [60]:
```
<AxesSubplot:xlabel='혈당', ylabel='총 콜레스테롤'>
```

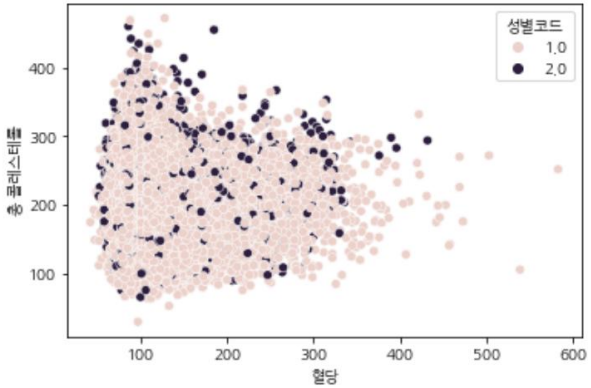


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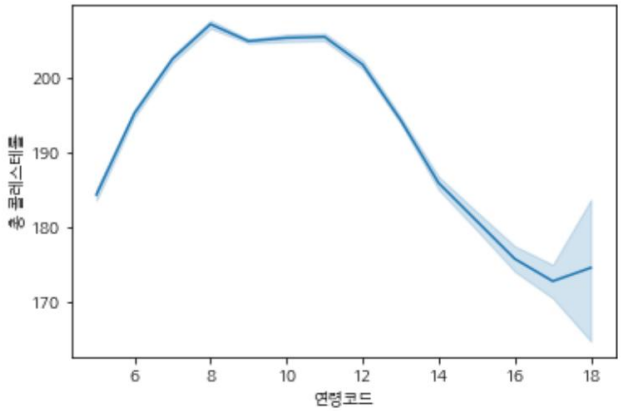
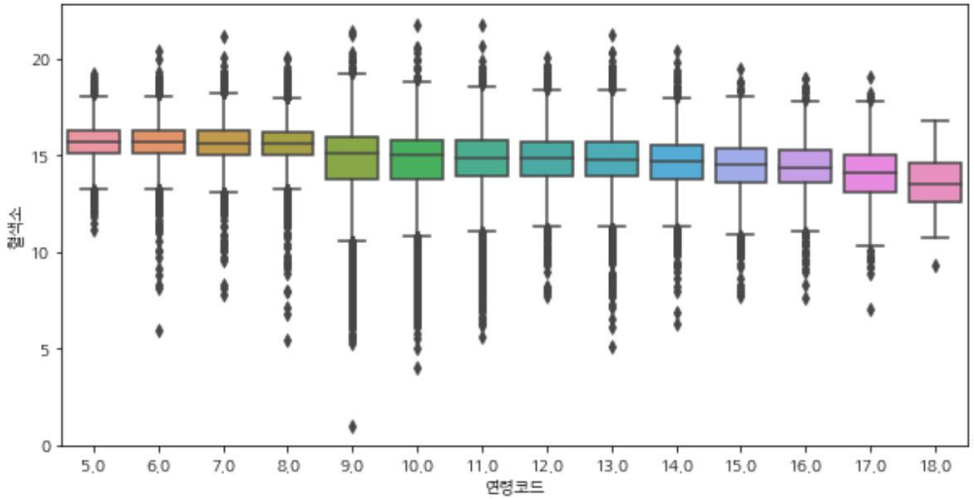
Chart creation practice (3)

ÿ Visual analysis and verification of various attribute values, such as trends, distributions, differences, and correlations between attributes

```
In [61]: sns.lineplot(x=df['연령코드'], y=df['총 콜레스테롤'])
Out[61]: <AxesSubplot:xlabel='연령코드', ylabel='총 콜레스테롤'>
```




```
In [62]: fig = plt.figure(figsize=(10,5))
         sns.boxplot(x=df['연령코드'], y=df['혈색소'])
Out[62]: <AxesSubplot:xlabel='연령코드', ylabel='혈색소'>
```